

Are Americans Getting Better at Surviving Extreme Weather?

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<https://wzhang76.github.io/econ238-portfolio/>

Extreme Weather Fatalities in the United States

- I used official data from the National Weather Service (NWS) to examine U.S. fatalities from tornadoes, floods, and hurricanes since 1940. The NWS data are compiled from Storm Data, which collects information from National Weather Service offices across the United States.
- For each year, I added the number of tornado, flood, and hurricane fatalities together. I then grouped the annual totals by decade and calculated the average number of fatalities per year for each period. This follows the assignment's request to report annual-average extreme-weather deaths by decade.

Period	Avg. Tornado Fatalities	Avg. Flood Fatalities	Avg. Hurricane Fatalities	Avg. Combined Fatalities per Year
1940–1949	178.8	61.9	21.6	262.3
1950–1959	140.9	79.1	87.7	307.7
1960–1969	93.5	129.7	58.7	281.9
1970–1979	98.6	181.9	21.7	302.2
1980–1989	52.1	109.7	11.8	173.6
1990–1999	57.9	99.2	14.0	171.1
2000–2009	55.8	64.7	115.4	235.9
2010–2019	91.0	98.8	5.3	195.1
2020–2024*	68.2	103.0	46.6	217.8

- For example, the average for the 1940s was calculated by adding tornado, flood, and hurricane fatalities for each year from 1940 through 1949 and then dividing the ten-year total by 10. The result was an average of 262.3 fatalities per year.
- The 2020–2024 period is shown separately because it is only a partial decade, so it should not be treated as directly equivalent to the completed ten-year periods above.

Source:

National Weather Service, 80-Year List of Severe Weather Fatalities, 1940–2024. The table reports annual tornado, flood, and hurricane fatalities used in the calculations above.

https://www.weather.gov/media/hazstat/80year_2024.pdf

Average Fatalities Over the Entire Period

- Using the same National Weather Service data from Part A, there were a total of 20,387 fatalities from tornadoes, floods, and hurricanes in the United States between 1940 and 2024. Since this period covers 85 years, I calculated the average as:

$$20,387 \div 85 = 239.85$$

- Therefore, the average number of fatalities from these three types of extreme weather was approximately 240 deaths per year over the entire 1940–2024 period.

Trend in Weather-Related Fatalities

- Looking only at tornado, flood, and hurricane deaths, I would say the general trend has gone down, but definitely not in a straight line. In the earlier decades, the average was usually around 260 to 300 deaths per year. By the 1980s and 1990s, it had dropped to around 170 deaths per year. There are still some periods where the number goes back up, so I would not say the decline is perfectly consistent.
- Once I include heat and cold deaths, though, the picture gets more complicated. The NWS heat and cold data start in 1986, and heat deaths especially can change a lot from year to year. More recently, NWS recorded 350 heat deaths in 2020, 375 in 2021, 383 in 2022, 555 in 2023, and 529 in 2024.
- So I would not simply say that weather-related deaths are falling. A better way to put it is that deaths from tornadoes, floods, and hurricanes have generally become lower over time, but once heat and cold are included, the overall trend is much less clear.

Population-Adjusted Death Intensity

- The raw death numbers are useful, but the U.S. population is a lot larger now than it was in the 1940s. Because of that, I decided to use deaths per million people as my “death intensity” measure. This way, a more accurate answer can be obtained.
- I calculated it as:
Death intensity = average annual deaths ÷ average population (in millions)

Period	Average U.S. Population (millions)	Avg. Annual Deaths	Deaths per Million People
1940–1949	137.7	262.3	1.91
1950–1959	163.9	307.7	1.88
1960–1969	191.6	281.9	1.47
1970–1979	214.4	302.2	1.41
1980–1989	237.0	173.6	0.73
1990–1999	264.5	171.1	0.65
2000–2009	294.6	235.9	0.80
2010–2019	319.2	195.1	0.61
2020–2024*	334.9	217.8	0.65

- My biggest discovery is that once the population is taken into account, this decline seems more obvious. For instance, the original death toll in the 1950s was actually higher than that in the 1940s, but the mortality rate per million people was slightly lower. By the 2010s, the mortality rate was only about 0.61 per million people, while it was approximately 1.91 per million people in the 1940s. This is not a completely smooth recession. In the first decade of this century, this ratio rose again, and the period from 2020 to 2024 is not even a complete decade. Despite this, after this survey, we can more clearly see that the probability of ordinary people being exposed to these three extreme weather risks is much lower than the original total number of deaths.

Sources:

National Weather Service, Weather Related Fatality and Injury Statistics: <https://www.weather.gov/hazstat>
U.S. Census Bureau, Historical and Annual Population Estimates: <https://www.census.gov/data/tables/time-series/demo/popest/pre-1980-national.html>
U.S. Census Bureau, National Population Totals, 2020s: <https://www.census.gov/data/datasets/time-series/demo/popest/2020s-national-total.html>

Are Americans Moving Toward Hurricane Risk?

- For this part, I looked at hurricanes. I found a study that compared county-level migration in the United States from 2010 to 2020 with natural hazard risk. The migration data came from the U.S. Census Bureau, and the hurricane risk data came from FEMA.
- What I find rather interesting is that this result is not as simple and straightforward as I initially thought. On the surface, regions with a higher risk of hurricanes actually have a greater inflow of population. This phenomenon indicates that places like Florida and Texas, although they have a higher risk of hurricanes, also have other more attractive aspects that attract more attention. For instance, job opportunities, housing, climate and lifestyle, etc. It seems that people give priority to these factors when moving compared to the risk of hurricanes.
- However, when the researchers controlled these factors, the results changed very obviously. That is to say, if the variables such as work, income and population density are excluded and only the item of “hurricane risk” is considered, it actually makes people less willing to move in.
- Therefore, this study indicates that one cannot simply say that people do not care about risks just because the population in high-risk areas is still growing. Often, people choose to move there because of other benefits rather than because they ignore the risk of hurricanes. These two are not in conflict.

Moving to Higher Ground

- For this part, I assumed that large buildings last around 50 years. I also assumed that moving or rebuilding them in a safer place would cost about 10% more than replacing them in the same place. These are just rough estimates, but I think they are reasonable enough for this calculation.
- The assignment says that about \$600 billion of capital is in hurricane-risk areas. If these buildings are replaced over 50 years, then around:
$$\$600 \text{ billion} \div 50 = \$12 \text{ billion}$$
would be replaced each year.
- If moving them to a safer place adds about 10% to the cost, then the extra cost would be:
$$\$12 \text{ billion} \times 10\% = \$1.2 \text{ billion per year}$$
If U.S. GDP is around \$31 trillion, then \$1.2 billion is only about 0.004% of GDP.
- For me, the main point is that this sounds much smaller than I expected. If buildings are moved in batches gradually when they need to be replaced, perhaps the cost will be much easier than moving all the buildings at once. Of course, this is a rough estimate, as the actual relocation will definitely involve other costs.

Disaster Losses

- From the report of SREX, I think the main reason for the increase in disaster losses is not merely that the weather itself may be changing. There are also other factors, such as more people, houses and other valuable things being located in areas that might be hit by the storm.
- Another important thing is how well a place is prepared. Two places may suffer from similar storms, but the damage caused may vary greatly depending on the buildings, infrastructure, income and the number of people affected.
- Therefore, I won't immediately say that the storm will definitely get worse just because I see higher disaster losses. Part of the reason for the increase might be that we have more people and property in dangerous areas.

Source:

IPCC, Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation (SREX), 2012. <https://www.ipcc.ch/report/managing-the-risks-of-extreme-events-and-disasters-to-advance-climate-change-adaptation/>

Heat-Related Deaths

- Heat deaths do not follow the same clear downward trend as tornado, flood, and hurricane deaths. The NWS recorded 1,021 heat deaths in 1995, and recent years have also remained high.
- The 1995 Chicago heatwave shows that temperature alone is not the whole story. Older people and those without air conditioning were especially at risk.
- This suggests that adaptation also matters. Air conditioning, warning systems, cool spaces, and checking on vulnerable people can all help reduce heat-related deaths.

Sources:

National Weather Service: https://www.weather.gov/media/hazstat/80year_2024.pdf
CDC, Heat-Related Mortality — Chicago, July 1995: <https://www.cdc.gov/mmwr/preview/mmwrhtml/00038443.htm>
CDC, Heat-Related Deaths — Chicago, Illinois, 1996–2001, and United States, 1979–1999: <https://www.cdc.gov/mmwr/preview/mmwrhtml/mm5226a2.htm>

Burke et al. (2025)

- Burke et al. find that cold causes more temperature-attributable mortality than heat, while also finding evidence of adaptation to local climates. Their estimates are different from direct NWS fatality counts.

But Wait...

- Lower storm mortality does not prove that storms themselves have become less dangerous. Better warnings, stronger buildings, infrastructure, and other forms of adaptation can also reduce deaths.